

LABLIST

**Contains all labs for the pro domains EXCEPT
FOR B.2.2 RIGHT NOW**

**CTRL + F first few words after “In this lab, your
task is to...”**

LAB NUMBER COMES AFTER

In this lab, your task is to:

- Change the password for the default pfSense account from pfsense to: **Donttre@donme**
- Create a new administrative user with the following parameters:
 - Username: **lyoung**
 - Password: **C@nyouGuessIt**
 - Full Name: **Liam Young**
 - Group Membership: **admins**
- Set a session timeout of **20** minutes for pfSense.
- Disable the webConfigurator anti-lockout rule for HTTP.

2.4.9

In this lab, your task is to:

- Perform a scan using the following information:
 - Network address: **73.44.216.0**
 - Subnet mask: **Class C**
- Answer the questions.

5.2.10

In this lab, your task is to use the Terminal to:

- Use **hciconfig** to discover and enable the onboard Bluetooth adapter.
- Use **hcitool** to find all of the Bluetooth devices.
- Answer Question 1.
- Use **l2ping** to determine if the Bluetooth device is alive and within range.
- Answer Question 2.
- Use **sdptool** to query Francisco's laptop to determine the Bluetooth services available on the device.
- Answer Question 3.
- Use **hcitool** to determine the clock offset and class for Brian's Braven Speaker device.
- Answer Question 4.

2.4.7

In this lab, your task is to:

- Use the **systemctl** command to start *bluetooth.service*.
- Use the **systemctl** command to stop *bluetooth.service*.
- Use the **systemctl** command to restart *bluetooth.service*.

4.1.5

In this lab, your task is to perform reconnaissance on the PartnerCorp external servers to find potentially vulnerable ports as follows:

On Analyst-Lap

- Use the **whois.org** website to determine the domain name servers used by PartnerCorp.xyz.
 - Answer Question 1.
- Use **nslookup** to determine the primary web server address.
 - Answer Question 2.

On Analyst-Lap2

- Use **Zenmap** to search for 50 of the top ports opened on the network identified by nslookup above.
 - Answer Question 3.

5.1.5

You are the security analyst for a small corporate network. While conducting some tests to see if your network can be hacked, you have discovered that you can obtain a copy of the zone information from the CorpDC3 server. This server is a domain controller in the CorpNet.local domain and holds an Active Directory-integrated zone for the CorpNet.local domain. To better protect the company zone data, you have decided to prevent zone transfers.

In this lab, your task is to disable zone transfers for the CorpNet.local zone.

6.1.11

In this lab, your task is to:

- Access the pfSense management console:
 - Username: **admin**
 - Password: **P@ssw0rd** (zero)
- Add a captive portal zone named **WiFi-Guest**.
 - Use the description **Guest wireless access zone**
- Using the GuestWi-Fi interface, configure your portal as follows:
 - Allow a maximum of **50** concurrent connections.
 - Disconnect user from the internet if their connection is inactive for **15** minutes.
 - Disconnect user from the internet after **45** minutes regardless of their activity.

- Limit users' downloads and uploads to **7000** and **2400** Kbit/s, respectively.
- Force to pass through your portal prior to authentication.
- Allow the following MAC and IP address to pass through the portal:
 - MAC: **00:00:1C:11:22:33**
 - IP: **198.28.1.100/16**
 - Give the IP address the description of **Security analyst's laptop**

2.4.6

You are the security analyst for a small corporate network. You want to ensure that your local network is free from potential vulnerabilities. You have decided to get a list of the operating systems running on your network and to see if any folders are shared.

In this lab, your task is to complete the following:

- From IT-Laptop, use Zenmap to determine the operating system of the hosts on your network.
 - The network address is **192.168.0.0/24**
 - Answer Question 1.
- From ITAdmin and Windows PowerShell (Admin), complete the steps below in order:
 - Use **net view** to check for shared folders on the CorpFiles and CorpFiles16 servers.
 - Map the H: drive to the shared folder found on CorpFiles.
 - View the files in the shared folder on the CorpFiles server.
 - Answer Question 2.

Launch Lab

5.3.10

In this lab, your task is to:

- Sign in to pfSense using the following case-sensitive information:

- URL: **198.28.56.22**
- Username: **admin**
- Password: **pfsense**
- Configure the DNS servers as follows:
 - Primary DNS server: **163.128.78.93** - Hostname: **DNS1**
 - Secondary DNS server: **163.128.80.93** - Hostname: **DNS2**
- Configure the WAN IPv4 information as follows:
 - Enable the interface.
 - Use a static IPv4 address of **65.86.24.136/8**
 - Add a new gateway using the following information:
 - Type: **Default gateway**
 - Name: **WANGateway**
 - IP address: **65.86.1.1**

2.4.9

In this lab, your task is to:

- Use **nmap** to perform a port scan on the database server to determine if any ports are open.
- Answer the question.

5.1.10

In this lab, your task is to:

- Create a honeypot on the computer named **www_stage** using Pentbox.
- Using Google Chrome, test the honeypot on the computer named Marketing3 using the **www_stage.corpnet.xyz** URL.
- Using the **www_stage** system, review the effects of the intrusion.
- Answer the questions.

3.4.10

In this lab, your task is to complete the following:

- From Office2 (the golden system), create an XML file that will be used to create the initial code integrity policy (CIPolicy).
 - When running the command:
 - Use the **-Level Pca** and **-UserPEs** flags.
 - Scan the entire C:\ drive.
- Convert the XML file to a binary file named **AppCIP.bin**.
- Save the converted file in the CorpDC\WDAC share.

7.3.4

In this lab, your task is to configure Windows Security as follows:

- Add a file exclusion for **D:\Graphics\book.jpg**.
- Add a process exclusion for files associated with your corporate software (**corp_process.exe**).
- Check for virus and threat updates.
- Perform a quick scan.

7.3.3

In this lab, your task is to:

- Use Zenmap to run a scan on the **192.168.0.0/24** network to look for the following open ports:
 - Port 3389 - Remote Desktop Services (TermServices)
 - Port 5900 - VNC Server (vncserver)
 - Answer Questions 1 and 2.
- Disable and stop the services for the open ports found running on the applicable computers.
Use the following table to identify the computers:

4.1.7

In this lab, your task is to:

- Use the **systemctl** command to enable *anaconda.service*.
- Use the **systemctl** command to disable *vmtoolsd.service*.
- After each command, check the service status with the **systemctl is-enabled** command.

Launch Lab

4.1.6

In this lab, your task is to use **nmap** to detect open ports as follows:

- Scan the computers in Building A using **192.168.0.0/24** for the network address.
- Scan the computers in Building B using **192.168.10.0/24** for the network address.
- Scan the company's public-facing web servers using **198.28.1.0/24** for the network address.
- Answer the questions.

5.2.11

In this lab, your task is to:

- Run the **firewall-bypass** nmap script against the firewall.
- Answer the question.

5.2.13

In this lab, your task is to:

- Use the **auxiliary/scanner/mssql/mssql_ping** exploit in Metasploit to determine which TCP port Microsoft SQL is using.
- Answer the question.

5.3.12

In this lab, your task is to:

- Use the post/windows/gather/enum_patches exploit in Metasploit to enumerate the Windows patches that are missing or vulnerable.
- Answer the question.

5.3.11

PRODOMAIN B.2.3 END^^^^

In this lab, your task is to:

- Set a secure passcode on the iPad as follows:
 - Require a passcode: **After 5 minutes**
 - New passcode: **youwontguessthison**
- Turn simple passcodes off.
- Configure the iPad to erase data after 10 failed passcode attempts.

Launch Lab

2.4.8

In this lab, your task is to:

- Use Zenmap to find which ports on CorpServer (192.168.0.10) are open.
- Use Wireshark and the enp2s0 network interface to determine if the CorpServer is under a SYN attack.
- Analyze the packets captured.
- Answer the questions.

5.1.15

In this lab, your task is to:

- Use Wireshark to capture packets and filter for packets with the SYN flag set.
 - Filter for SYN packets with **tcp.flags.syn==1 and tcp.flags.ack==0**
- Use a filter to display only packets that have the SYN flag and the ACK flag set.
 - Filter for SYN-ACK packets with **tcp.flags.syn==1 and tcp.flags.ack==1**
- Use a filter to only display packets that contain the ACK flag and answer the question.
 - Filter for ACK packets with **tcp.flags.syn==0 and tcp.flags.ack==1**
 - Answer the question.

6.9.12

In this lab, your task is to create and configure a network policy to allow members of the sales department to remotely connect using the following parameters:

- Name the policy **Sales**.
- Use a remote access server.
- Connecting users/computers must belong to the **Sales** group.
- Deny access to any account that is not configured in Active Directory.
- Configure permissions to use settings in the Active Directory user accounts (User Dial-in properties). User account settings are configured by an Active Directory user account administrator.
- Use the secured password (EAP-MSCHAP v2).
- Configure a session timeout of **30** minutes.
- As a constraint, allow access only from 6:00 a.m. to 9:00 p.m., Monday–Friday.
- Make the policy first in the list of policies.

7.1.4

In this lab, your task is to determine whether ARP poisoning is taking place by doing the following:

- Use Wireshark to capture packets on the enp2s0 interface for five seconds.
- Analyze the Wireshark packets to determine whether ARP poisoning is taking place.

- Use the 192.168.0.2 IP address to help make your determination.
- Answer the questions.

6.5.9

In this lab, your task is to:

- View normal access to the RMK Office Supplies website (**rmksupplies.com**).
- Use Ettercap to:
 - Begin unified sniffing on the enp2s0 interface.
 - Set Exec (192.168.0.30) as the target machine.
 - Add and enable DNS spoofing using an Ettercap plugin.
 - Initiate ARP poisoning on remote connections.
- From Exec, use Google Chrome to access **rmksupplies.com** and analyze the results.
- Answer the question.

6.5.10

In this lab, your task is to capture and analyze a TCP SYN flood attack as follows:

- Use Wireshark to capture packets on the enp2s0 interface.
- Using the Linux Terminal, ping CorpTest (192.168.10.19) to verify connectivity. Note the packets in Wireshark.
- Set a display filter in Wireshark to show only TCP SYN packets.
- Use **hping3** to launch a SYN flood attack against CorpTest (192.168.10.19) .
- After capturing packets for a few seconds, examine a SYN packet with the destination address of 192.168.10.19.
- Answer the question.

6.9.11

In this lab, your task is to:

- Scan all devices using an IP address in the range of 192.168.200.50 through 192.168.200.100.
- Answer the questions.

5.2.14

In this lab, your task is to scan for rogue wireless access points from a terminal as follows:

- Use **airmon-ng** to discover and enable the onboard wireless adapter.
- Use **airodump-ng** to scan for wireless access points.
- Answer the questions.

6.2.8

PRO DOMAIN B.2.4 END^^^^

In this lab, your task is to configure the following UAC settings in the Default Domain Policy on CorpDC:

User Account Control	Setting
Admin Approval Mode for the Built-in Administrator account	Enabled
Allow UIAccess applications to prompt for elevation without using the secure desktop	Disabled
Behavior of the elevation prompt for administrators in Admin Approval mode	Prompt for credentials
Behavior of the elevation prompt for standard users	Automatically deny elevation requests
Detect application installations and prompt for elevation	Enabled
Only elevate executables that are signed and validated	Disabled

Only elevate UIAccess applications that are installed in secure locations	Enabled
Run all administrators in Admin Approval Mode	Enabled
Switch to the secure desktop when prompting for elevation	Enabled
Virtualize file and registry write failures to per-user locations	Enabled

7.2.14

The password requirements for your website are as follows:

- The password must be eight or more characters in length.
- The password must include at least one upper and one lowercase letter.
- The password must have at least one of these special characters: !, ", #, \$, %, &, _, ', *, or @
- All passwords are encrypted using an md5 or sha1 hash algorithm.

2.5.5

In this lab, your task is to configure BitLocker drive encryption as follows:

- Try to turn on BitLocker for the System (C:) drive.
- Answer the question.
- From the BIOS settings, turn on and activate TPM.
- Turn on BitLocker for the System (C:) drive.
- Save the recovery key to **\\CorpServer\BU-Exec**.
- Encrypt the entire **System (C:)** drive.
- Run the BitLocker system check.

7.1.15

In this lab, your task is to use the Local Security Policy tool to configure password restrictions as follows:

- New passwords cannot be the same as the previous 4 passwords.
- Passwords must be changed every 30 days.
- New passwords cannot be changed for at least 2 days.
- Passwords must be at least 10 characters long.
- Passwords must contain non-alphabetical characters.
- Automatically unlock locked accounts after 1 hour.
- Lock the user account after four incorrect logon attempts within a 40-minute period.

2.5.6

In this lab, your task is to complete the following:

- Approve the pending certificate request for a smart card certificate from mlopez.
- Deny the pending web server certificate request for CorpSrv16.
- User bnguyen lost his smartcard. Revoke the certificate assigned to bnguyen.CorpNet.com using the Key Compromise reason code.
- User tsutton has left his company. Revoke the certificate assigned to tsutton.CorpNet.com using the Change of Affiliation reason code.
- Unrevoke the CorpDev2 certificate.

2.5.7

In this lab, your task is to:

- Install the smart card key readers in the appropriate locations to control access to key infrastructure.
- Install the IP security cameras in the appropriate locations to record which employees access the key infrastructure.
- Install a Restricted Access sign in the appropriate location to control access to the key infrastructure.
- Add the visitor log to a location appropriate for logging visitor access.

2.3.4

PRODOMAIN5 B25 END^^^

In this lab, your task is to complete the following:

- On IT-Laptop, use Ettercap to launch an on-path DHCP spoofing attack using the following parameters:
 - Netmask: **255.255.255.0**
 - DNS Server IP: **192.168.0.11**
 - View the IP address.
 - Answer Question 1.
- On Support, complete the following tasks:
 - Using Wireshark and the enp2s0 interface, start a capture and filter the display, showing only DHCP traffic.
 - From the Terminal, view the default gateway addresses.
 - Answer Question 2.
 - Bring the network interface down and back up to request new DHCP addresses.
 - View the gateway again to see if anything has changed.
 - Answer Question 3.
 - In Wireshark, analyze the captured DHCP ACK packets to find the router IP addresses that were used.
 - Answer Question 4.
- On Office1, analyze the effects of the on-path attack by completing the following tasks:
 - View the default gateway.
 - Answer Question 5.
 - Use **tracert** to find the path to rmksupplies.com.
 - Answer Question 6.
 - Release and renew the IP addresses assigned by DHCP.
 - Check the default gateway of the computer again. What has changed?
 - Answer Question 7.

- Use **tracert** to find the path to rmksupplies.com. Has anything changed?
- Answer Question 8.
- Log in to the rmksupplies.com Employee Portal with the following credentials:
 - Username: **bpascal**
(Blaise Pascal)
 - Password: **Dontrygu3ss1ng**
- On IT-Laptop, find the captured username and password in Ettercap.

6.6.6

In this lab, your task is to complete the following:

- On ITAdmin, use the Security Evaluator (found on the taskbar) to run a vulnerability scan.
- On Office2, remediate all vulnerabilities that didn't pass as shown in the vulnerability report.
- On ITAdmin, re-run a vulnerability scan to make sure all of the issues are resolved.

Launch Lab

5.4.12

In this lab, your task is to:

- Access the pfSense management console:
 - Username: **admin**
 - Password: **P@ssw0rd** (zero)
- Create and configure a firewall rule to pass HTTP traffic from the WAN to the web server in the screened subnet.
- Create and configure a firewall rule to pass HTTPS traffic from the WAN to the web server in the screened subnet.
 - Use the following table when creating the HTTP and HTTPS firewall rules:

6.1.13

In this lab, your task is to complete the following:

- On IT-Laptop, use **airmon-ng** to put the wireless adapter in monitor mode.
- Use **airodump-ng** to find the hidden access point.
- Answer the question.
- From the Gst-Lap computer, connect to the rogue AP using the SSID of BookStore.

Launch Lab

6.2.6

In this lab, your task is to use Burp Suite to evaluate website logins as follows:

- Open Burp Suite and turn Intercept off.
- Monitor the HTTP History tab while logging in to the rmksupplies.com Employee Portal.
 - Username: **sramirez**
 - Password: **mickeyminniegoofypluto**
- Examine the POST entry in HTTP History to discover if this website is vulnerable to attack.
- Answer the question.

5.4.15

In this lab, your task is to analyze an Office1 disk image with Autopsy by doing the following:

- Create a case file.
 - Case Name: **corpnet-case132** (no spaces)
 - Base Directory: **d:\autopsy**
 - Case Number: **132**
 - Examiner: enter your *name*, *number*, and *email*
- Import the disk image.
 - Disk image file: **e:\office1_hd.vhd**
 - Ingest Modules: **Recent Activity**, **Hash Lookup**, **File Type Identification**, **EXIF Parser**, and **Keyword Search**
- Analyze the disk image.
 - Under Data Sources, analyze the Users folder on vol2.

- Answer Question 1.
- Under Views, analyze the File Types and the File Size.
- Answer Questions 2 and 3.
- Under Results, analyze the Extracted Content and the Keyword Hits.
- Answer Questions 4-6.

7.1.17

In this lab, your task is to use **ping** and **hping3** to troubleshoot connectivity for the guest servers.

- Use **ping** and **hping3** to test connectivity to each server and note the results.

Server	IP Address
CorpDC	192.168.0.11
CorpFiles16	192.168.0.12
CorpFiles	192.168.0.13
CorpDHCP	192.168.0.14
CorpWeb	192.168.0.15
CorpDC2	192.168.0.16
CorpWSUS1	192.168.0.17
CorpCA_Card	192.168.0.18

- Answer the questions.

5.2.15

In this lab, your task is to use pfSense's Snort to complete the following:

Sign into pfSense using the following:

- Username: **admin**
- Password: **P@ssw0rd** (zero)
- Download the:
 - Snort free registered User rules
 - Oinkmaster Code: **992acca37a4dbd7**
 - Snort GPLv2 Community rules
 - Emerging Threats Open rules
 - Sourcefire OpenAppID detectors
 - APPID Open rules
- Configure rule updates to happen every **4** days at **12:10 a.m.**
 - Hide any deprecated rules.
- Block offending hosts for 1 day.
- Send all alerts to the system log when the Snort starts and stops.
- Assign Snort to the WAN interface using a description of **Snort-WAN**
 - Include:
 - Sending alerts to the system log
 - Automatically blocking hosts that generate a Snort alert
- Start Snort on the WAN interface.

6.1.12

You are the security analyst for a small corporate network. Several employees have reported that they are unable to connect to the network. They all seem to be getting bad IP address information from a rogue DHCP server.

In this lab, your task is to identify the rogue DHCP server using Wireshark:

- Use Wireshark to capture and filter DHCP traffic.
- Disable and enable the enp2s0 network interface to request a new IP address from DHCP.
- Find the rogue and legitimate DHCP servers.

- Answer the questions.

Use **bootp** in Wireshark to isolate DHCP traffic.

6.2.7

In this lab, your task is to complete the following:

- Use Zenmap to scan for open ports running VNC. Use the table below to help you identify the computer.
- Go to the suspect computer and uninstall VNC.

If needed for Linux systems, use **dnf list vnc** to view package names and **dnf erase package_name** to uninstall the package.

- From the suspect computer, run **netstat -l** to verify that the ports for VNC are closed.

IP Address	Computer
192.168.0.30	Exec
192.168.0.31	ITAdmin
192.168.0.32	Gst-Lap
192.168.0.33	Office1
192.168.0.34	Office2
192.168.0.46	IT-Laptop
192.168.0.47	Support

5.2.12

In this lab, your task is to:

- Sign in to pfSense using:

- Username: admin
- Password: P@ssw0rd (zero)
- Create a firewall rule that blocks all DNS traffic leaving the LAN network.
- Create a firewall rule that allows all DNS traffic going to the LAN network.

Use the following table for the two rules:

Parameter	Setting
Protocol	UDP (53)
Descriptions	For the block rule: Block DNS from LAN For the allow rule: Allow all DNS to LAN

- Arrange the firewall rules in the order that allows them to function properly.
- Enable and configure pfBlockerNG using the information in the following table:

7.3.6

You are the security analyst for a small corporate network. To ensure that your Linux computers meet the current company security standards, you have decided to conduct a vulnerability scan.

In this lab, your task is to complete the following:

- From the ITAdmin computer, use the Security Evaluator to check the security:
 - On the Linux computer with the IP address of 192.168.0.22
 - On the Linux computers in the IP address range of 192.168.0.60 through 192.168.0.69
- Answer the questions.

5.4.13

B.2.2 END